# LORAINE SANDOVAL ASCENCIO

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## Education

**UNIVERSITY OF CALIFORNIA, IRVINE** — Irvine, CA

**Doctor of Philosophy in Physics with Concentration in Astronomy and Astrophysics** — June 2027

**CALIFORNIA STATE UNIVERSITY, LOS ANGELES** — Los Angeles, CA

**Master of Science in Physics** — May 2021

**UNIVERSITY OF CALIFORNIA, LOS ANGELES** — Los Angeles, CA

**Bachelor of Science in Astrophysics** — June 2019

## Research Experience

**GRADUATE STUDENT RESEARCHER** — Irvine, CA

**University of California, Irvine** — June 2022 – Present
Advisor: Dr. Michael C. Cooper

- Plan and conduct observing runs with the Keck Cosmic Web Imager (KCWI), including selecting and observing targets for spectroscopic follow-up.
- Process raw observational data using the KCWI data reduction pipeline for low-surface brightness galaxies.
- Extract and analyze 1D spectra for 45 candidate ultra-diffuse galaxies, including measurements of redshifts and stellar-population properties.
- Analyze the star-formation histories, atomic gas content, and environments of ultra-diffuse galaxies to investigate the physical mechanisms driving their evolution and quenching.

**GRADUATE STUDENT RESEARCHER** — Irvine, CA

**University of California, Irvine** — Aug. 2021 – June 2022
Advisor: Dr. Asantha Cooray

- Simulated galaxy spectra with the Bagpipes modeling tool, incorporating the resolution of the upcoming SPHEREx NASA survey to assess observational capabilities.
- Investigated how variations in galaxy parameters (e.g., star formation history, metallicity) influence the spectral features SPHEREx will detect, informing survey design and analysis strategies.

**INTERSTELLAR AND HELIOSPHERIC PHYSICS GROUP INTERN** — Pasadena, CA

**Internship at NASA's Jet Propulsion Laboratory** — June 2020 – Aug. 2021

Advisors: Dr. Susan Terebey and Dr. Neal Turner

- Modeled and understood the physics of shocked gas from supersonic flow to determine how material enters the protostellar disk of young stellar objects and arrives at the protostar.
- Updated an existing code's (RadChemT) current model of gas dynamics in the protostellar disk of young stellar object L1527 for the molecule $C^{18}O$.

**RISE MS-TO-PHD FELLOW** — Los Angeles, CA

**California State University, Los Angeles** — June 2019 – Aug. 2021

Advisor: Dr. Susan Terebey

- Innovated new dynamics to be incorporated into an existing model for the young stellar object L1527.
- Researched the methods used to model the gas dynamics and kinematics in protostellar disks.

**COSMOLOGY RESEARCHER** Los Angeles, CA

**University of California, Los Angeles** Feb. 2019 – Aug. 2019

Advisor: Dr. Steven Furlanetto

- Developed a Python-based model to simulate ionizing radiation from O and B stars and predict recombination line emission in theoretical nebulae.
- Utilized the Cloudy photoionization software to analyze line intensity ratios (e.g., OII, OIII, SIII) and refine model parameters for emission nebulae.

**ORIGIN OF THE UNIVERSE GROUP INTERN** Pasadena, CA

**Internship at NASA's Jet Propulsion Laboratory** Sept. 2016 – Aug. 2017

Advisors: Dr. Graça Rocha & Dr. Krzysztof M. Gorski

- Extracted a temperature CMB angular power spectrum using 2015 Planck mission data and IDL algorithms available for conducting likelihood estimates.
- Replicated previously published results by Planck mission scientists in *Planck 2015 results XI. CMB power spectra, likelihoods, and robustness of parameters*.

**MANUFACTURING ENGINEERING INTERN** Pasadena, CA

**Internship at NASA's Jet Propulsion Laboratory** June 2016 – Aug. 2016

Advisors: Michael Verdin & Gustavo Maldonado

- Aided thorough research and compiled data on a list of electronic components, analyzing characteristics such as termination finish, hermeticity, and moisture sensitivity levels.
- Used 3D modeling tools to simulate a diode endcap and investigate solder fillet formation.

## Publications

- **Sandoval Ascencio, L.**, Cooper, M. C., Zaritsky, D., et al. 2026, in prep.
  *Gas-Rich Post-Starburst Ultra-Diffuse Galaxies: Quenching or Rejuvenation?*
- Khim, D. J., Zaritsky, D., **Sandoval Ascencio, L.**, et al. 2026, *ApJ*, submitted
  *SMUDGes Below the Surface: Clusters Confirm the Diversity of Low-Surface-Brightness Galaxies*
- Zaritsky, D., Donnerstein, R., **Sandoval Ascencio, L.**, et al. 2026, *ApJ,* 1006, 48
  *Systematically Measuring Ultradiffuse Galaxies. IX. A Gyr in the Life of Nearby Low Surface Brightness Galaxies*
- **Sandoval Ascencio, L.**, Cooper, M. C., Zaritsky, D., et al. 2025, *OJAp*, 8, 110
  *Caught in the Act of Quenching? — A Population of Post-Starburst Ultra-Diffuse Galaxies*
- Terebey, S., **Sandoval Ascencio, L.**, Flores-Rivera, L., et al. 2025, *ApJ*, 990, 53
  *The Dynamics of Infall and Accretion Shocks in the Outer Disk*
- Khim, D. J., Zaritsky, D., **Sandoval Ascencio, L.**, et al. 2025, *ApJ*, 989, 154
  *A Closer Look at an Unusual Ultradiffuse Galaxy*

## Teaching Experience

**TEACHING ASSISTANT** Irvine, CA

**University of California, Irvine** June 2022 – Present

- Lead discussion sections for both lower and upper-division undergraduate level physics and astronomy courses.
- Design worksheets to succinctly review each week's lecture material.
- Host office hours to provide homework help by effectively communicating examples.

**TEACHING ASSOCIATE (INSTRUCTOR, ONLINE)** Irvine, CA

**University of California, Irvine** June 2025 – July 2025

- Condensed a 10-week introductory astronomy curriculum into a 5-week accelerated online format while maintaining learning outcomes.
- Designed and recorded lecture videos (3+ per week) integrating interactive concept-check questions to promote active learning.
- Facilitated asynchronous discussions and provided timely feedback to support remote student engagement in a compressed schedule.

**TEACHING ASSOCIATE (INSTRUCTOR, IN PERSON)** Irvine, CA

**University of California, Irvine** September 2024 – December 2024

- Crafted and delivered engaging introductory astronomy lectures three times per week for 80+ undergraduate students.
- Integrated polling activities into lectures to encourage critical thinking and assess real-time understanding of key concepts.
- Developed and implemented qualitative and quantitative assessments to measure student performance.
- Collaborated with the teaching assistant to design effective assignments aligned with course objectives.

**SCIENCE TUTOR** Los Angeles, CA

**UCLA BruinCorps** Oct. 2017 – June 2018

- Assisted 30+ high school and middle school students with classwork and homework in a classroom setting.
- Supervised 30+ students during work periods.
- In-class tutored by proactively checking in with students.

**MATH & SCIENCE TUTOR** Sylmar, CA

**LAMC STEM Math Center** Sept. 2014 – Aug. 2016

- Assisted 20+ college students in targeted subject areas in a group setting.
- In-class tutored during classroom exercises and proactively checked in with students.
- Facilitated workshops on selected topics in physics and math by effectively communicating examples.

## Professional Service

**GALFRESCA 2026 ORGANIZING COMMITTEE MEMBER** Irvine, CA

**University of California, Irvine** Sept 2026

- Co-organized the 2026 Galaxy Formation & Evolution in Southern California (GalFRESCA) conference, bringing together more than 60 researchers from institutions across Southern California.
- Coordinated conference programming and logistics, including website building, scheduling talks, and communicating with participants.

**SACNAS NATIONAL LIAISON** Irvine, CA

**University of California, Irvine** Sept. 2026 – June 2027

- Co-advise the graduate and undergraduate boards and foster collaborations with chapters across campuses.
- Track chapter events and milestones for annual reporting to SACNAS National.

**SACNAS GRADUATE PRESIDENT** Irvine, CA

**University of California, Irvine** Sept. 2025 – June 2026

- Led the Graduate Executive Board in supporting underrepresented students in STEM through community building, professional development, and outreach.

- Oversaw academic, social, and outreach programming and coordinated collaborations with campus and UC partners.
- Mentored undergraduate board members and strengthened partnerships with campus organizations.

**SACNAS SOCIAL CHAIR** Irvine, CA

**University of California, Irvine** Sept. 2024 – June 2025

- Organized inclusive social and networking events to foster community among SACNAS members.
- Mentored undergraduate social coordinators and collaborated with campus organizations on joint events.

**PHYSICS GRADUATE CAUCUS SOCIAL EVENTS CHAIR** Irvine, CA

**University of California, Irvine** Sept. 2022 – June 2024

- Organized academic and social events to foster community among physics graduate students.
- Increased student engagement through initiatives such as themed socials and stress-relief activities.

**PHYSICS AND ASTRONOMY COMMUNITY EXCELLENCE (PACE) MENTOR** Irvine, CA

**University of California, Irvine** Sept. 2022 – June 2023

- Peer mentored a first-year graduate student in the Physics and Astronomy Department.
- Met biquarterly for in-person meetings to discuss future research plans and offer advice for finding an advisor.

**RISING STAR GIRLS TEAM MEMBER** Irvine, CA

**University of California, Irvine** Summers 2022 – 2026

- Assisted in preparation for a future week-long workshop designed to encourage underrepresented middle school girls to explore the sciences.
- Led workshop activities as a facilitator in an effort to combine both art and astronomy.
- Translated documents (registration, media release, etc.) for Spanish-speaking parents or guardians.

**TRANSFER STUDENT MENTOR** Los Angeles, CA

**UCLA Transfer Center** Sept. 2018 – June 2019

- Guided new transfer students in their transition to UCLA by sharing knowledge based on previous experience.
- Acquired information for new transfer students by attending mentor workshops.
- Built community among a cohort of 5 students through regular meetings and communication.

**PRESIDENT OF ASTRONOMY CLUB** Los Angeles, CA

**LA Mission College Astronomy Club** Sept. 2016 – June 2017

- Planned stargazing events, guest speakers, and trips with the help of board members.
- Organized and led biweekly meetings.

## Academic Conferences

- National Society for Black Physicists – National Society for Hispanic Physicists Joint Conference, Nov. 2026
- NDiSTEM Conference, Oct. 2025, 2026
- Keck Science Meeting, Sept. 2025, 2026
- Galaxy Formation & Evolution in Southern California (GalFRESCA) Conference, Sept. 2024, 2025, & 2026
- 10 Years of Ultra-Diffuse Galaxies: Past Perspectives and Future Prospects, Aug. 2025
- Sierra Conference for UC Astro Grads, Sept. 2024
- 236$^{th}$ Meeting of the American Astronomical Society (AAS), June 2020 (remote)
- Cal State LA Women in Data Science (WiDS) Conference, Mar. 2020
- 28$^{th}$ Annual Cal State LA Student Symposium on Research, Scholarship and Creative activities, Feb. 2020

## Invited Talks and Community Engagement

- Lunch Seminar Speaker for the Department of Astrophysical and Planetary Science at CU Boulder, Nov. 2026
- Session Panelist at the NDiSTEM (SACNAS) Conference, Oct. 2026
- Solar System Exhibit UCI Volunteer at the OC Discovery Cube, Mar. 2025 & 2026
- Science Demonstrations with SACNAS at the Downtown Santa Ana Artwalk, Dec. 2025
- SACNAS at UCI Transfer Visit Day Panel, May 2025
- Los Angeles Mission College STEM Alumni Panel, Mar. 2025
- Magnolia Public Schools STEAM Expo Judge, April 2024
- Magnolia Public Schools Science Fair Demonstration for Students, Mar. 2024
- Cal State LA Departmental Colloquium, Aug. 2020
- John F. Kennedy High School College Panel, Mar. 2019
- LA Mission College STEM Coffee Hour, Sept. 2018
- UCLA Women and Youth Supporting Each Other (WYSE) Empowerment Day Student Panel, Apr. 2018
- LA Mission College Society of Hispanic Professional Engineers (SHPE) Student Panel, Mar. 2017

## Professional Development & Training

- CSU Pre-Professor Program (PREPP) Fellow, CSU Long Beach, Spring 2026
- Code/Astro software development workshop at Northwestern University, July 2023
- Observational Astronomy Workshop at UCO/Lick Observatory, May 2023

## Certifications

- Summer Teaching Apprenticeship Program (STAP), Aug. 2025
- Mentoring Excellence Program (MEP), Feb. 2025
- Activate to Captivate Public Speaking, Mar. 2024
- Industry Insights for STEM Scientists, Nov. 2023

## Fellowships & Scholarships

- José Flores Velázquez Endowment in Astrophysics, University of California, Irvine, June 2025
- Graduate Opportunity Fellowship, University of California, Irvine, 2021-2022
- Research Initiative for Scientific Enhancement MS-to-PhD Fellowship, CSU Los Angeles, Minority Opportunities in Research Programs, 2019-2021
- UCLA Regents Scholarship, 2017-2019

## Honors & Awards

- Graduate Oral Presentation Awardee at the NDiSTEM (SACNAS) Conference, Oct. 2026
- UCI Latino Excellence and Achievements (LEAD) Award for Graduate Student Excellence, April 2026
- California State University, Los Angeles Special Recognition in Graduate Studies, May 2021
- UCLA Dean’s Honor List, Winter 2019 & Spring 2019
- UCLA Transfer Mentorship Program Certificate of Achievement, 2019
- LA Mission College President’s Honor List, Spring 2017
- LA Mission College Dean's Honor List, Spring 2017, Fall 2016, Spring 2016, Spring 2015, & Fall 2014

## Skills & Abilities

**DATA HANDLING**

- Processing, analyzing, modeling, and managing large-scale astronomical datasets, including image reduction, spectral analysis, and multi-wavelength data integration

**COMPUTER PROGRAMMING LANGUAGES**

- Proficient in Python; experience with C, C++, and IDL

**LANGUAGE**

- Proficient in Spanish language